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Interconnect for IC package

Abstract

An integrated circuit (IC) package includes an interconnect comprising patches of unoxidized metal that are circumscribed by a region of roughened metal formed of oxidized metal. The IC package also includes a die mounted on the interconnect. The die is conductively coupled to at least a subset of the patches of unoxidized metal.

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Background/Summary

TECHNICAL FIELD

(1) This description relates to an interconnect for an integrated circuit (IC) package.

BACKGROUND

(2) An interconnect (alternatively referred to as a lead frame) is a metal structure inside an integrated circuit (IC) package that carries signals from a die of the IC package to external components. An interconnect includes a central die pad, where the die is placed, surrounded by pads situated on a periphery of the interconnect.

(3) Interconnects are manufactured by removing material from a flat plate of copper, copper-alloy, an iron nickel alloy, etc. To form the IC package, a die is adhered to the die pad of the interconnect. Also, wire bonds are attached between the die and pads of the interconnect to connect the die to the pads. In a process referred to as encapsulation, a plastic case is molded around the lead frame and die, exposing only the pads. The pads are cut off outside the plastic body and any exposed supporting structures are cut away.

SUMMARY

(4) A first example relates to an integrated circuit (IC) package that includes an interconnect with patches of unoxidized metal that are circumscribed by a region of roughened metal formed of oxidized metal. The IC package also includes a die mounted on the interconnect. The die is conductively coupled to at least a subset of the patches of unoxidized metal.

(5) A second example relates to a method for forming an IC package. The method includes selectively etching a resist overlaying a sheet of interconnects to provide a mask of resist on the sheet of interconnects. The method also includes roughing, in response to the selectively etching, the sheet of interconnects. The method includes removing a remaining portion of the resist to provide the sheet of interconnects with units that have a smoothed portion and a roughened portion.

The smoothed portion of the units has spaced apart patches of unoxidized metal, and the roughened portion has oxidized metal that circumscribed the patches of unoxidized metal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates a diagram of a cross-sectional view of an example of an integrated circuit (IC) package.
(2) FIG. **2** illustrates an overhead view of an interconnect of FIG. **1**.
(3) FIG. **3** illustrates a diagram of a cross-sectional view of another example of an IC package.
(4) FIG. **4** illustrates an overhead view of an interconnect of FIG. **3**.
(5) FIG. **5** illustrates a first stage of an example method for forming interconnects for IC packages.
(6) FIG. **6** illustrates a second stage of an example method for forming interconnects for IC packages.
(7) FIG. **7** illustrates a third stage of the example method for forming interconnects for IC packages.
(8) FIG. **8** illustrates a fourth stage of the example method for forming interconnects for IC packages.
(9) FIG. **9** illustrates a fifth stage of the example method for forming interconnects for IC packages.
(10) FIG. **10** illustrates a sixth stage of the example method for forming interconnects for IC packages.
(11) FIG. **11** illustrates a seventh stage of the example method for forming interconnects for IC packages.
(12) FIG. **12** illustrates a first stage of an example method for forming an IC package.
(13) FIG. **13** illustrates a second stage of the example method for forming an IC package.
(14) FIG. **14** illustrates a third stage of the example method for forming the IC package.
(15) FIG. **15** illustrates a first stage of another example method for forming interconnects for IC packages.
(16) FIG. **16** illustrates a second stage of the other example method for forming interconnects for IC packages.
(17) FIG. **17** illustrates a third stage of the other example method for forming interconnects for IC packages.
(18) FIG. **18** illustrates a fourth stage of the other example method for forming interconnects for IC packages.
(19) FIG. **19** illustrates a fifth stage of the other example method for forming interconnects for IC packages.
(20) FIG. **20** illustrates a sixth stage of the other example method for forming interconnects for IC packages.
(21) FIG. **21** illustrates a seventh stage of the other example method for forming interconnects for IC packages.
(22) FIG. **22** illustrates a first stage of another example method for forming an IC package.
(23) FIG. **23** illustrates a second stage of the other example method for forming an IC package.
(24) FIG. **24** illustrates a third stage of the other example method for forming the IC package.
(25) FIG. **25** illustrates a fourth stage of the other example method for forming the IC package.
(26) FIG. **26** illustrates a flowchart of an example method for processing a sheet of interconnects for IC packages.
(27) FIG. **27** illustrates a flowchart of an example method for forming an IC package.

DETAILED DESCRIPTION

(28) This description relates to IC packages and methods for forming the IC packages. The IC

packages include an interconnect that has a smoothed portion (unoxidized metal, such as unoxidized copper) and a roughened portion (oxidized metal, such as oxidized copper). More particularly, the interconnect includes spaced apart patches of unoxidized metal (part of the smoothed portion) that are circumscribed by a roughened region formed of oxidized metal (part of the roughened portion).

(29) In some examples, the die is attached to the interconnect with a flip chip process. In such examples, the die includes posts extending from a surface of the die to contact the spaced apart patches of unoxidized metal on the interconnect. The spaced apart patches of unoxidized metal on the interconnect allow for improved solder wetting to securely mount the die on the interconnect. In complementary fashion, the roughened region of the interconnect that circumscribes the spaced apart patches of unoxidized metal impedes the flow of solder, thereby preventing solder bridges between posts of the die.

(30) In other examples, the die is mounted on the interconnect with wire bonding. In such examples, the interconnect includes a die pad at a center of the interconnect, and pads at a periphery of the interconnect. In these examples, a first surface of the die is mountable on the die pad. The die is adhered to the die pad with an epoxy to electrically insulate the die from the die pad. Also, wire bonds are adhered to a second surface of the die, and to patches of the pads of the interconnect. The wire bonds are adhered to the patches of unoxidized metal on the pads of the interconnect with solder, and the patches of unoxidized metal provide improved solder wetting. In complementary fashion, the roughened region of the interconnect that circumscribes the spaced apart patches of unoxidized metal impedes the flow of solder, thereby preventing solder bridges between wire bonds.

(31) Additionally, a molding (e.g., a mold compound) encases the interconnect and the die to provide the IC package, such as a quad flat no leads (QFN) IC package or a dual flat no leads (DFN) IC package. The portion of roughened metal (e.g., oxidized copper) on the interconnect provides improved purchase (e.g., grip) for the molding, such that the molding is securely adhered to the interconnect. The patches of unoxidized metal (forming the portion of smoothed metal) provide improved solder wetting compared to a region of roughened metal. Complementarily, the roughened portion formed of oxidized metal impedes the flow of solder and provides improved adherence to the molding relative to the smoothed region. Accordingly, by providing the interconnect with both the smoothed portion (e.g., unoxidized metal) and the roughened portion (e.g., oxidized metal), the benefits of an interconnect formed with either unoxidized metal or roughened metal are provided without the corresponding hindrances.

(32) FIG. **1** illustrates a cross-sectional view of an integrated circuit (IC) package **100**. The IC package **100** includes a die **104** mounted on an interconnect **108** (e.g., a lead frame). In the example illustrated, the die **104** is mounted on the interconnect **108** using flip chip techniques to form a QFN or a DFN package. The interconnect **108** has a smoothed portion of metal (e.g., unoxidized copper) and a roughened portion of metal (e.g., oxidized metal, such as oxidized copper). In particular, the interconnect **108** includes a central region **112** that has a surface **116** that has a region of roughened metal **118** (oxidized metal) that is interrupted by spaced apart patches of unoxidized metal **120**. That is, the spaced apart patches of unoxidized metal **120**, which are a component of the smoothed portion of the interconnect **108** are circumscribed by the region of roughened metal **118** (e.g., oxidized metal), which is a component of the roughened portion of the metal. Stated differently, the smoothed portion of the interconnect **108** includes the spaced apart patches of unoxidized metal **120** that are circumscribed by part of the roughened portion of the interconnect, namely the region of roughened metal **118** of the central region **112**.

(33) The die **104** includes posts **124** that extend from a surface **128** of the die **104** to the spaced apart patches of unoxidized metal **120** on the central region **112** of the interconnect **108**. The posts **124** are adhered to the spaced apart patches of unoxidized metal **120** with solder **132** (e.g., a solder ball or a solder paste). The region of roughened metal **118** circumscribes the spaced apart patches

of unoxidized metal **120**. Thus, the region of roughened metal **118** forms a continuous surface on the central region **112** of the interconnect **108**, such that a point of the region of roughened metal **118** of the surface **116** of the central region **112** is connected to another point on the central region **112** through a path (linear or curved) without crossing one of the spaced apart patches of unoxidized metal **120**.

(34) FIG. **2** illustrates an overhead view **170** of the interconnect **108** of FIG. **1**, wherein the die **104** is removed. Thus, for purposes of simplification of explanation, FIGS. **1** and **2** employ the same reference numbers to denote the same structures. The interconnect **108** includes extensions **134** that extend from the central region **112** of the interconnect to a pad (hidden from view) of a first type arranged on a periphery of the interconnect **108**. The interconnect **108** also includes pads **136** of a second type arranged around a periphery of the interconnect **108**. The pads **136** of the second type include a patch of unoxidized metal **140** that is circumscribed by a remaining portion of roughened metal **144** (e.g., oxidized metal). That is, the pads **136** of the second type include a first portion of unoxidized metal, namely, the patch of unoxidized metal **140**, and a second (remaining) portion of oxidized metal, namely the region of roughened metal **144**. Thus, the patch of unoxidized metal **140** on the pads **136** of the second type is part of the smoothed portion of the interconnect **108**. Similarly, the region of roughened metal **144** on the pads **136** of the second type is part of the roughened portion of the interconnect **108**. Conversely, the pads **135** of the first type have a surface that is covered completely (or nearly completely) with roughened metal (hidden from view in FIG. **1**).

(35) In the example illustrated, there are six (6) spaced apart patches of unoxidized metal **120** on the central region **112** of the interconnect **108**, only some of which are labeled. In other examples, there are more or less spaced apart patches of unoxidized metal **120**. Additionally, there are four (4) pads **135** of the first type and eight (8) pads **136** of the second type, only some of which are labeled. In various examples, there are more or less pads **135** of the first type and more or less pads **136** of the second type. The spaced apart patches of unoxidized metal **120** are circumscribed by the region of roughened metal **118** of the central region **112**. Additionally, as illustrated in the overhead view **170**, the pads **136** of the second type are arranged at the periphery of the interconnect **108**. The pads **136** of the second type include the patch of unoxidized metal **140** circumscribed by the region of roughened metal **144**. Conversely, the pads **135** of the first type include a region of roughened metal **174** and do not include a patch of unoxidized metal.

(36) Referring back to FIG. **1**, the spaced apart patches of unoxidized metal **120** at the central region **112** of the interconnect **108** enable excellent solder wetting. Similarly, patches of unoxidized metal **140** on the pads **136** of the second type also enable excellent solder wetting. Accordingly, the spaced apart patches of unoxidized metal **120** and the patches of unoxidized metal **140** enable the posts **124** to be securely adhered to the spaced apart patches of unoxidized metal **120** of the central region **112** of the interconnect **108** and to the pads **136** of the second type. Accordingly, the die **104** is electrically coupled to the set of unoxidized metal patches, namely, the spaced patches of unoxidized metal **120** on the central region **112** and the patches of unoxidized metal **140** on the pads **136** of the second type. In complementary fashion, the region of roughened metal **118** on the central region **112** of the interconnect provides a barrier for the solder **132** to prevent solder bridging between posts **124** of the die **104**.

(37) The interconnect **108** and the die **104** are encased in a molding **160**. The molding **160** is formed of a molding compound, such as plastic or similar material. Also, the region of roughened metal **118** on the central region **112**, the region of roughened metal **144** on the pads **136** of the second type and the roughened metal **174** on the pads **135** of the first type improve adherence of the molding **160** to the interconnect **108**.

(38) By employment of the interconnect **108**, the benefits of roughing portions (oxidizing metal) of the interconnect **108** are attained without drawbacks. More particularly, because the interconnect **108** includes the spaced apart patches of unoxidized metal **120** (components of the smoothed

portion) on the central region **112** and the pads **136** of the second type include the patches of unoxidized metal **140**, the die **104** is securely mounted on the interconnect **108** to enable robust electrical communication between the die **104** and the pads **135** of the first type and the pads **136** of the second type. More specifically, the smoothed patches of unoxidized metal allow for excellent solder wetting. Complementarily, the interconnect **108** also includes the region of roughened metal **144** at the pads **136** of the second type (circumscribing the patches of unoxidized metal **140**), the region of roughened metal **118** of the central region **112** (circumscribing the spaced apart patches of unoxidized metal **120**) as well as the roughened metal **174** on the pads **135** of the first type. These regions of oxidize metal forming the region of roughened metal **118** of the central region **112** and the regions of roughened metal **144** on the pads **136** of the second type impede the flow of solder, thereby preventing solder bridges. Stated differently, the regions of oxidize metal forming the region of roughened metal **118** on the central region of the interconnect **108** provides a barrier to prevent solder adhering to a given post **124** to the central region **112** or a pad **136** of the second type from bridging to another post **124**. These regions of roughened metal, such as the region of roughened metal **118**, the roughened metal **174** on the pads **135** of the first type and roughened metal **144** on the pads **136** of the second type also improve purchase (e.g., grip) between the interconnect **108** and the molding **160**, thereby improving adherence between the molding **160** and the interconnect **108**. Accordingly, providing the interconnect **108** with patches of unoxidized metal (e.g., smoothed copper) and regions of roughened metal (e.g., oxidized copper), provides the benefits of an interconnect formed with an unoxidized metal and an interconnect formed with an unoxidized metal without the respective drawbacks.

(39) FIG. **3** illustrates a cross-sectional view of another example IC package **200**. The IC package **200** includes a die **204** mounted on an interconnect **208** (e.g., a lead frame) with wire bonding to form a QFN or DFN package. More particularly, the die **204** is mounted on a die pad **212** of the interconnect **208**. The interconnect **208** has a smoothed portion of metal (e.g., unoxidized copper) and a roughened portion of metal (e.g., oxidized metal, such as oxidized copper). In particular, the die pad **212** has a surface **216** that has a region of roughened metal **218** (oxidized metal) that is interrupted by a patch of unoxidized metal **220**. That is, the patch of unoxidized metal **220**, which is a component of the smoothed portion of the interconnect **208** is circumscribed by the region of roughened metal **218** (e.g., oxidized metal), which is a component of the roughened portion of the metal. Stated differently, the smoothed portion of the interconnect **208** includes the patch of unoxidized metal **220** that is circumscribed by part of the roughened portion of the interconnect **208**, namely the region of roughened metal **218** of the die pad **212**.

(40) A first surface **224** of the die **204** is mounted on the patch of unoxidized metal **220** with a layer of epoxy **228** (e.g., an industrial glue) to adhere the die **204** to the die pad **212** and to provide electrical isolation between the first surface **224** of the die **204** and the die pad **212**. That is, the first surface **224** of the die **204** is adhered to the patch of unoxidized metal **220** of the die pad **212** with the epoxy **228**. The region of roughened metal **218** circumscribes the patch of unoxidized metal **220**. Thus, the region of roughened metal **218** forms a continuous surface, such that a point of the region of roughened metal **218** of the surface **216** of the die pad **212** is connected to another point on the surface **216** through a path (linear or curved) without crossing the patch of unoxidized metal **220**.

(41) The interconnect **208** also includes pads **236** arranged around a periphery of the interconnect **208**. The pads **236** include a patch of unoxidized metal **240** that is circumscribed by a remaining portion of roughened metal **244** (e.g., oxidized metal). That is, the pads **236** include a first portion of unoxidized metal, namely, the patch of unoxidized metal **240**, and a second (remaining) portion of oxidized metal, namely the region of roughened metal **244**. Thus, the patch of unoxidized metal **240** on the pads **236** is part of the smoothed portion of the interconnect **208**. Similarly, the region of roughened metal **244** on the pads **236** is part of the roughened portion of the interconnect **208**.

(42) FIG. **4** illustrates an overhead view **270** of the interconnect **208** of FIG. **3**, wherein the die **204**

is removed. Thus for purposes of simplification of explanation, FIGS. **3** and **4** employ the same reference numbers to denote the same structures. As illustrated, there are twelve (12) patches of unoxidized metal **240** on the pads **236**. In other examples, there are more or less pads **236** such that there are corresponding more or less patches of unoxidized metal **240** on the pads **236**. The patches of unoxidized metal **220** are circumscribed by the region of roughened metal **218** of the die pad **212**. Additionally, as illustrated in the overhead view **270**, the pads **236** are arranged at the periphery of the interconnect **208**. The pads **236** include the patch of unoxidized metal **240** circumscribed by the region of roughened metal **244**.

(43) Referring back to FIG. **3**, wire bonds **248** are coupled to a second surface **250** of the die **204** and to the patch of unoxidized metal **240** on the pads **236** through solder **252** (e.g., a solder ball or solder paste). Accordingly, the die **204** is electrically coupled to the patches of unoxidized metal **240**. The second surface **250** of the die **204** opposes the first surface **224** of the die **204**. Because the die **204** is electrically insulated from the patch of unoxidized metal **220**, the die **204** is electrically coupled to a subset of patches of unoxidized metal, namely the patches of unoxidized metal **240**.

(44) The patches of unoxidized metal **220** of the die pad **212** enables excellent epoxy wetting. Accordingly, the patch of unoxidized metal **220** allows the die **204** to be securely adhered to the die pad **212**. Similarly, the patch of unoxidized metal **240** on the pads **236** enables excellent solder wetting to allow the solder **252** to securely adhere to the pads **236**. Additionally, the region of roughened metal **244** circumscribing the corresponding patch of unoxidized metal **240** prevents the solder from bridging with another connection.

(45) The interconnect **208** and the die **204** are encased in a molding **260**. The molding **260** is formed of a molding compound, such as plastic or similar material. The region of roughened metal **244** on the pads **236** and the region of roughened metal **218** of the die pad **212** improve adherence of the molding **260** to the interconnect **208**.

(46) By employment of the interconnect **208**, the benefits of roughing portions (oxidizing metal) of the interconnect **208** are attained without drawbacks. More particularly, because the die pad **212** includes the patch of unoxidized metal **220** (components of the smoothed portion) and the pads **236** include the patches of unoxidized metal **240**, the die **204** is securely mounted on the interconnect **208** to enable robust electrical communication between the die **204** and the pads **236**. More specifically, the smoothed patches of unoxidized metal allow for excellent solder wetting. Complementarily, the interconnect **208** also includes the region of roughened metal **244** at the pads **236** (circumscribing the patches of unoxidized metal **240**) and the region of roughened metal **218** of the die pad **212** (circumscribing the patch of unoxidized metal **220**). The regions of oxidize metal forming the region of roughened metal **218** of the die pad **212** and the roughened metal **244** of the pads **236** provides a barrier to prevent solder bridging between connections. These regions of roughened metal also improve purchase (e.g., grip) between the interconnect **208** and the molding **260**, thereby improving adherence between the molding **260** and the interconnect **208**. Accordingly, providing the interconnect **208** with patches of unoxidized metal (e.g., smoothed copper) and regions of roughened metal (e.g., oxidized copper), provides the benefits of an interconnect formed with an unoxidized metal and an interconnect formed with an unoxidized metal without the respective drawbacks.

(47) FIGS. **5-11** illustrate methods for forming interconnects for an IC package, such as the IC package **100** of FIG. **1**. For purposes of simplification of explanation, FIGS. **5-11** employ the same reference numbers to denote the same structure.

(48) In a first stage, as illustrated in FIG. **5**, at **300**, a sheet of interconnects **400** is provided. In some examples, the sheet of interconnects **400** is formed of copper (Cu). In other examples, other oxidizable metals are employable to form the sheet of interconnects. The sheet of interconnects include units **402** that are singulatable into individual interconnects, such as the interconnect **108** of FIGS. **1-2**. These units **402** include a central region **404**, pads **406** of a first type and pads **408** of a

second type arranged around a periphery of each respective unit **402** in the sheet of interconnects **400**. Only some of the pads **406** of the first type and pads **408** of the second type are labeled.
(49) In a second stage, as illustrated in FIG. **6**, at **305**, the sheet of interconnects **400** is pretreated with industrial cleaning material to clean the sheet of interconnects **400**. In a third stage, as illustrated in FIG. **7**, at **310**, a layer of resist **412** (e.g., photoresist) is laminated (e.g., deposited) on the sheet of interconnects **400**. In some examples, the layer of resist **412** is a coating of polyimide overlaying a surface of the sheet of interconnects **400**.
(50) In a fourth stage, as illustrated in FIG. **8** at **315**, the resist **412** is selectively etched to pattern the sheet of interconnects **400** to expose portions of the sheet of metal (e.g., bare copper). Stated differently, at **315**, the resist **412** is selectively etched to provide a mask of resist **412** on the sheet of interconnects **400**. More particularly, the resist **412** is etched such that the resist (the resist **412** of FIG. **7**) covers portions of the sheet of interconnects **400** that will form the central regions **404** and pads **406** of the interconnects, and remaining portions (only some of which are labeled) of the sheet of interconnects **400** are exposed metal. Additionally, a region **416** of the resist **412** is not etched to allow formation of a barcode on the sheet of interconnects **400**.
(51) In a fifth stage, as illustrated in FIG. **9** at **320**, the exposed portions of the sheet of interconnects **400** are roughened. More particularly, the remaining portion of the sheet of interconnects **400** that is exposed (at operation **315**) is oxidized to roughen a surface of the sheet of interconnects **400** (formed of metal) to provide a portion of roughened metal **418** of the sheet of interconnects **400**. Additionally, the portions of the sheet of interconnects **400** underlying the resist **412** remain unoxidized (e.g., remain smooth). In a sixth stage, as illustrated in FIG. **10**, at **325**, the remaining portions of the resist **412** are removed (e.g., in a wet etching process) to reveal a sheet of interconnects **400** that each have a central region **404** and pads **406** of the first type and pads **408** of the second type. The sheet of interconnects **400** includes patches of unoxidized metal **422**, only some of which are labeled. The patches of unoxidized metal **422** are employable to receive posts of a die on a central region **404** that are circumscribed by a region of roughened metal **418** (oxidized portion of metal). Further, at **325**, a barcode **428** is inscribed (e.g., with a laser) on a patch of unoxidized metal corresponding to the region **416** of the resist of FIG. **9**. In some examples, the barcode **428** is a two-dimensional barcode (e.g., a quick response (QR) code). In other examples, the barcode **428** is a one-dimensional barcode. The barcode **428** allows the sheet of interconnects **400** to be uniquely identified in a database. Further, by inscribing the barcode **428** on the patch of unoxidized metal **416**, the readability of the barcode **428** is improved. Conversely, if a barcode were to be inscribed in the region of roughened metal **418**, such a barcode would have an inconsistent surface (due to the roughing), which would make such a barcode difficult to read. Conversely, the barcode **428** is inscribed in unoxidized (smoothed) metal, enabling a scanner (or other device) to read the barcode **428** without difficulty.
(52) In a seventh stage, as illustrated in FIG. **11**, at **330**, the units **402** in the sheet of interconnects **400** of FIG. **10** are singulated to provide an array of interconnects **440**. The singulation can be executed, for example, with a laser, a saw, bending, stretching, etc. The singulation separates the barcode **428** from the array of interconnects **440**. Each resultant interconnect **440** includes a central region **404**, pads **406** of the first type and pads **408** of the second type. The central region **404** of the interconnects **440** and the pads **408** of the second type include patches of unoxidized metal **422** for mounting posts. Additionally, the pads **406** of the first type of the interconnects **440** are covered with roughened metal. Each interconnect **440** is employable to implement the interconnect **108** of FIGS. **1** and **2**.
(53) FIGS. **12-14** illustrate methods for forming an IC package using flip chip techniques, such as the IC package **100** of FIG. **1**. For purposes of simplification of explanation, FIGS. **12-14** employ the same reference numbers to denote the same structure.
(54) In a first stage, as illustrated in FIG. **12**, at **500**, an interconnect **600** is provided. In some examples, the interconnect **600** is formed from the method of FIGS. **5-11**. The interconnect **600**

includes a central region **604** situated at a center of the interconnect **600**. Also, the interconnect **600** includes pads **608** (e.g., pads **136** of the first type of FIG. **1**) arranged at a periphery of the interconnect **600**. The central region **604** includes spaced apart patches of unoxidized metal **612** (smoothed regions) that are circumscribed by a region of roughened metal **616** (oxidized metal). Similarly, the pads **608** include patches of unoxidized metal **620** circumscribed by regions of roughened metal **624**.

(55) In a second stage, as illustrated in FIG. **13**, at **505**, a die **630** is mounted on the interconnect **600**. The die **630** includes posts **634** extending from a surface **638** of the die **630** to a corresponding patch of unoxidized metal **612**. The posts **634** are mounted on the spaced apart patches of unoxidized metal **612** with solder **642**, such as solder past or a solder ball. The region of roughened metal **616** impedes the flow of solder. Thus, the solder **642** for adhering each respective post **634** to a corresponding patch of unoxidized metal **612** is contained within or near the boundaries of the patch of unoxidized metal **612**, such that solder bridges between posts **634** are not inadvertently formed.

(56) In a third stage, as illustrated in FIG. **14**, at **510**, a molding **662** (e.g., a molding compound) encases the interconnect **600** and the die **630** to form the IC package **666**. The regions of roughened metal **624** on the pads **608** and the region of roughened metal **616** on the central region **604** ensure that the molding **662** is securely adhered to the interconnect **600**. Accordingly, as described herein, providing the interconnect **600** with patches of unoxidized metal (e.g., smoothed copper) and regions of roughened metal (e.g., oxidized copper), provides the benefits of an interconnect formed of an oxidized metal and an interconnect formed of an unoxidized metal without the respective drawbacks.

(57) FIGS. **15-21** illustrate another example method for forming interconnects for an IC package, such as the IC package **200** of FIG. **3**. For purposes of simplification of explanation, FIGS. **15-21** employ the same reference numbers to denote the same structure.

(58) In a first stage, as illustrated in FIG. **15**, at **700**, a sheet of interconnects **800** is provided. In some examples, the sheet of interconnects **800** is formed of copper (Cu). In other examples, other oxidizable metals are employable to form the sheet of interconnects. The sheet of interconnects include units **802** that are singulatable into individual interconnects, such as the interconnect **208** of FIGS. **3-4**. These units **802** include a die pad **804** and pads **808** arranged around a periphery of each respective unit **802** in the sheet of interconnects **800**. Only some of the pads **808** are labeled.

(59) In a second stage, as illustrated in FIG. **16**, at **705**, the sheet of interconnects **800** is pretreated with industrial cleaning material to clean the sheet of interconnects **800**. In a third stage, as illustrated in FIG. **17**, at **710**, a layer of resist **812** (e.g., photoresist) is laminated (e.g., deposited) on the sheet of interconnects **800**. In some examples, the layer of resist **812** is a coating of polyimide overlaying a surface of the sheet of interconnects **800**.

(60) In a fourth stage, as illustrated in FIG. **18** at **715**, the resist **812** is selectively etched to pattern the sheet of interconnects **800** to expose portions of the sheet of metal (e.g., bare copper). Stated differently, at **315**, the resist **812** is selectively etched to provide a mask of resist **812** on the sheet of interconnects **800**. More particularly, the resist **812** is etched such that the resist (the resist **812** of FIG. **17**) covers portions of the sheet of interconnects **800** that will form the die pads **804** and pads **808** of the interconnects, and remaining portions (only some of which are labeled) of the sheet of interconnects **800** are exposed metal. Additionally, a region **816** of the resist **812** is not etched to allow formation of a barcode on the sheet of interconnects **800**.

(61) In a fifth stage, as illustrated in FIG. **19** at **720**, the exposed portions of the sheet of interconnects **800** are roughened. More particularly, the remaining portion of the sheet of interconnects **800** that is exposed (at operation **315**) is oxidized to roughen a surface of the sheet of interconnects **800** (formed of metal) to provide a portion of roughened metal **818** of the sheet of interconnects **800**. Additionally, the portions of the sheet of interconnects **800** underlying the resist **812** remain unoxidized (e.g., remain smooth). In a sixth stage, as illustrated in FIG. **20**, at **725**, the

remaining portions of the resist **812** are removed (e.g., in a wet etching process) to reveal a sheet of interconnects **800** that each have a die pad **804** and pads **808** of the second type. The sheet of interconnects **800** includes patches of unoxidized metal **822**, only some of which are labeled. The patches of unoxidized metal **822** on the die pad **804** are employable to mount a die, and the patches of unoxidized metal **822** on the pads **808** are employable for attaching a wire bond to couple the die and the pads **808**. The patches of unoxidized metal **822** are circumscribed by a region of roughened metal **818** (oxidized portion of metal). Further, at **325**, a barcode **828** is inscribed (e.g., with a laser) on a patch of unoxidized metal corresponding to the region **816** of the resist of FIG. **19**. In some examples, the barcode **828** is a two-dimensional barcode (e.g., a QR code). In other examples, the barcode **828** is a one-dimensional barcode. The barcode **828** allows the sheet of interconnects **800** to be uniquely identified in a database. Further, by inscribing the barcode **828** on the patch of unoxidized material, the readability of the barcode **828** is improved. Conversely, if a barcode were to be inscribed in the region of roughened metal **818**, such a barcode would have an inconsistent surface (due to the roughing), which would make such a barcode difficult to read. In contrast, the barcode **828** is inscribed in unoxidized (smoothed) metal, enabling a scanner (or other device) to read the barcode **828** without difficulty.

(62) In a seventh stage, as illustrated in FIG. **21**, at **730**, the units **802** in the sheet of interconnects **800** of FIG. **20** are singulated to provide an array of interconnects **840**. The singulation can be executed, for example, with a laser, a saw, bending, stretching, etc. The singulation separates the barcode **828** from the array of interconnects **840**. Each resultant interconnect **840** includes a die pad **804** and pads **808**. The die pad **804** of the interconnects **840** and the pads **808** include patches of unoxidized metal **822** for mounting posts. Each interconnect **840** is employable to implement the interconnect **208** of FIGS. **3** and **4**.

(63) FIGS. **22-25** illustrate methods for forming an IC package using wire bonding, such as the IC package **200** of FIG. **3**. For purposes of simplification of explanation, FIGS. **22-25** employ the same reference numbers to denote the same structure.

(64) In a first stage, as illustrated in FIG. **22**, at **900**, an interconnect **1000** is provided. In some examples, the interconnect **1000** is formed from the method of FIGS. **15-21**. The interconnect **1000** includes a die pad **1004** situated at a center of the interconnect **1000**. Also, the interconnect **1000** includes pads **1008** arranged at a periphery of the interconnect **1000**. The die pad **1004** includes a patch of unoxidized metal **1012** (smoothed regions) on a surface **1016** that is circumscribed by a region of roughened metal **1018** (oxidized metal). Similarly, the pads **1008** include patches of unoxidized metal **1020** circumscribed by regions of roughened metal **1024**.

(65) In a second stage, as illustrated in FIG. **23**, at **905**, a die **1030** is mounted on the die pad **1004** of the interconnect **1000**. More particularly, a first surface **1034** of the die **1030** is mounted on the patch of unoxidized metal **1012** with an epoxy **1038**. The region of roughened metal **1018** impedes the flow of epoxy **1038**.

(66) In a third stage, as illustrated in FIG. **24** at **910**, wire bonds **1050** are mounted on a second side **1054** of the die **1030** and on a respective pad **1008**. The wire bonds **1050** are mounted on the pads **1008** with solder **1058** (e.g., a solder ball or solder paste). The region of roughened metal **1024** on each pad **1008** impedes the flow of solder, such that the solder **1058** is contained within or near boundaries of the patch of unoxidized metal **1020**.

(67) In a fourth stage, as illustrated in FIG. **25**, at **915**, a molding **1062** (e.g., a molding compound) encases the interconnect **1000** and the die **1030** to form the IC package **1066**. The regions of roughened metal **1024** on the pads **1008** and the region of roughened metal **1018** on the die pad **1004** ensure that the molding **1062** is securely adhered to the interconnect **1000**. Accordingly, as described herein, providing the interconnect **1000** with patches of unoxidized metal (e.g., smoothed copper) and regions of roughened metal (e.g., oxidized copper), provides the benefits of an interconnect formed of an oxidized metal and an interconnect formed of an unoxidized metal without the respective drawbacks.

(68) FIG. **26** illustrates a flowchart of an example method **1100** for providing interconnects for forming IC packages, such as the IC package **100** of FIG. **1** or the IC package **200** of FIG. **2**. At **1105**, a sheet of interconnects (e.g., the sheet of interconnects **400** of FIG. **5** or the sheet of interconnects **800** of FIG. **15**) is provided.
(69) The sheet of interconnects includes units that are singulatable into individual interconnects. The sheet of interconnects is formed, for example, with copper. At **1110**, the sheet of interconnects is pretreated in a cleaning process.
(70) At **1115**, a layer of resist (e.g., the resist **412** of FIG. **7**) is deposited on the sheet of interconnects, such that the resist overlays the sheet of interconnects. At **1120**, the resist is selectively etched to provide a mask on the sheet of interconnects, such as illustrated in FIG. **8**. More specifically, the resist is etched such that patches (e.g., the patches of resist **412** of FIG. **8**) of the resist are circumscribed by exposed bare metal.
(71) At **1125**, in response to the selectively etching, the sheet of interconnects are roughened such that the exposed bare metal is oxidized. Conversely, portions of the sheet of interconnects covered by the patches of resist remain unoxidized.
(72) At **1130**, a remaining portion of the resist (e.g., the patches of resist) is removed to provide a sheet of interconnects comprising a units that have a smoothed portion of unoxidized metal and a roughened portion of metal. The smoothed portion of the units are spaced apart patches of unoxidized metal on a die pad and a patch of unoxidized metal on pads of the units. At **1135**, a barcode (e.g., the barcode **428** of FIG. **10**) is inscribed (e.g., with a laser) in a particular patch of unoxidized metal.
(73) At **1140**, the units in the sheet of interconnects are singulated to provide individual interconnects. The units are singulated, for example, by lasing, sawing, stretching and/or bending. At **1145**, IC packages are formed using some or all of the (singulated) interconnects.
(74) FIG. **27** illustrates a flowchart of an example method **1200** for fabricating an IC package with an interconnect (e.g., the interconnect **108** of FIG. **1** or the interconnect **208** of FIG. **3**). In some examples, the IC package is a QFN IC package. In other examples, the IC package is a DFN IC package. The method **1200** is employable to implement an instance of the operation at **1145** of FIG. **26** (e.g., as a sub-process). In some examples, the interconnect includes a central region (e.g., the central region **112** of FIG. **1**) with spaced apart patches of unoxidized metal (e.g., unoxidized copper) In other examples, the interconnect includes a die pad. Additionally, the interconnect includes pads (e.g., the pads **136** of FIG. **1**) that have a patch of unoxidized metal. A remaining portion of the die pad and the pads is oxidized metal (e.g., oxidized copper).
(75) At **1205**, a die (e.g., the die **104** of FIG. **1**) is mounted on the interconnect. In some examples, such as examples where the die is mounted using flip chip techniques, the die includes posts that extend from a surface of the die. In this example, the posts are situated to contact the patches of unoxidized metal and are adhered to such patches with solder. In other examples, such as examples where the die is mounted using wire bonding, a first surface of the die is mounted on a die pad and adhered to a patch of the unoxidized metal on the die pad with an epoxy.
(76) At **1210**, in some examples, wire bonds are attached to a second surface of the die and to the patches of unoxidized metal on the pads of the interconnect. The second surface of the die opposes the first surface of the die. In other examples, the operations at **1210** are omitted. At **1215**, a molding is applied to encase the die and the interconnect in the molding (e.g., the molding **160** of FIG. **1**) to provide the IC package. The roughened portion of the interconnect (e.g., the portions of oxidized metal) enable improved purchase (e.g., grip) for the molding to adhere to the interconnect.
(77) Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. An integrated circuit (IC) package comprising: an interconnect comprising patches of unoxidized metal that are circumscribed by a region of roughened metal formed of oxidized metal; pads arranged at a periphery of the interconnect, wherein the pads comprise a patch of the patches of unoxidized metal circumscribed by a portion of the region of roughened metal; and a die mounted on the interconnect, wherein the die is conductively coupled to at least a subset of the patches of unoxidized metal.
2. The IC package of claim 1, wherein the unoxidized metal is copper and the oxidized metal is oxidized copper.
3. The IC package of claim 1, wherein the die comprises posts that extend to the spaced apart patches of unoxidized metal.
4. The IC package of claim 3, wherein the region of roughened metal provides a barrier to prevent solder adhering to one of the posts from bridging to others of the posts.
5. The IC package of claim 1, further comprising: a molding that encases the die and the interconnect.
6. The IC package of claim 1, wherein the IC package is a quad flat no leads (QFN) IC package.
7. An integrated circuit (IC) package comprising: pads adjacent to and at a periphery of a die, wherein the pads each comprise a patch of unoxidized metal circumscribed by a portion of a region of roughened metal; and a die mounted on the patches of unoxidized metal, there being no gap between a side surface of at least one of the patches of unoxidized metal and a side surface of the region of roughened metal formed of oxidized metal, wherein the die is conductively coupled to at least a subset of the patches of unoxidized metal.
8. The IC package of claim 7, wherein the unoxidized metal is copper and the oxidized metal is oxidized copper.
9. The IC package of claim 7, wherein the die comprises posts that extend to the spaced apart patches of unoxidized metal.
10. The IC package of claim 9, wherein the region of roughened metal provides a barrier to prevent solder adhering to one of the posts from bridging to others of the posts.
11. The IC package of claim 7, further comprising: a molding that encases the die and at least portions of the patches of unoxidized metal that are circumscribed by the region of roughened metal formed of oxidized metal.
12. The IC package of claim 7, wherein the IC package is a quad flat no leads (QFN) IC package.
13. An integrated circuit (IC) package comprising: a die attach pad formed of oxidized metal; pads spaced from a periphery of the die attach pad, at least some of the pads comprising a patch of unoxidized metal circumscribed by a region of oxidized metal; a die mounted on the die attach pad; and wherein the die is conductively coupled to at least a subset of the patches of the pads.
14. The IC package of claim 13, wherein the die comprises posts that extend to the pads spaced from the periphery of the die attach pad.
15. The IC package of claim 14, wherein the oxidized metal provides a barrier to prevent solder adhering to a given post of the posts to the die pad from bridging to another post of the posts.
16. The IC package of claim 13, further comprising: a molding that encases the die and at least portions of the die attach pad and the pads spaced from the periphery of the die attach pad.
17. The IC package of claim 13, wherein the IC package is a quad flat no leads (QFN) IC package.
18. An integrated circuit (IC) package comprising: a die attach pad including a patch of unoxidized metal circumscribed by a region of roughened metal formed of oxidized metal; a die mounted on the patch of unoxidized metal; and wherein at least some of the pads include patches of unoxidized metal that are circumscribed by a region of roughened metal formed of oxidized metal, there being no gap between a side surface of at least one of the patches of unoxidized metal and a side surface of the region of roughened metal formed of oxidized metal, wherein the die is conductively coupled to at least a subset of the patches of unoxidized metal.

19. The IC package of claim 18, further including pads spaced from the periphery of the die attach pad.
20. The IC package of claim 19, wherein the die comprises bond pads electrically coupled to the pads spaced from the periphery of the die attach pad.
21. The IC package of claim 19, further comprising: a molding that encases the die and covers at least portions of the die attach pad and the pads spaced from the periphery of the die attach pad.
22. The IC package of claim 21, wherein the IC package is a quad flat no leads (QFN) IC package.
23. The IC package of claim 18, wherein the die comprises bond pads electrically coupled to the pads spaced from the periphery of the die attach pad.